

A Polynomial Fold with a Nondefinable Non-Archimedean Wave-Front Set

Run `fa_banach_001`, lane 10

July 22, 2026

Status. Candidate counterexample, likely valid, subject to human review. The result answers the broad $\mathcal{C}^{\text{exp}}$ -class reading of the source question. A narrower question with a definable input density is not settled.

1 Source question

Cluckers, Halupczok, Loeser, and Raibaut ask in Example 3.4.6 of [1], on PDF page 35, whether wave-front sets of special distributions are definable; their examples expressly include pushforwards of smooth distributions under polynomial maps.

So far, we know by Theorem [3.4.1](#) that they are zero loci of functions of $\mathcal{C}^{\text{exp}}$ -class (with conic structure).

Also, it would be interesting, for wave front sets of special distributions, to check whether they are definable sets, like for push-forwards under polynomial mappings of smooth distributions, their Fourier transforms, etc. Note that the wave front set of a push-forward of a smooth distribution under a polynomial mapping f is contained in N_f by Theorem [2.9.6](#) and that N_f itself is a definable set.

The surrounding paper works with uniform families of $\mathcal{C}^{\text{exp}}$ -class distributions over non-archimedean local fields. We use “definable” in its generalized Denef–Pas sense, as in [1].

2 Construction and statement

Let F be a non-archimedean local field of odd residue characteristic, with valuation $\text{ord} : F^\times \rightarrow \mathbb{Z}$, valuation ring $\mathcal{O}_F$, and residue-field cardinality q_F . Haar measure is normalized by $\text{vol}(\mathcal{O}_F) = 1$. Put

$$X_F = (\mathcal{O}_F \setminus \{0\})^2$$

and, for $a = (x, y) \in X_F$, define

$$h_F(a) = q_F^{\text{ord } x} - \text{ord } y.$$

The function h_F is locally constant on X_F . Uniformly in F , it is a constructible, hence $\mathcal{C}^{\text{exp}}$ -class, function: the formalism permits integer-valued definable functions and generators $q_F^{\alpha(a)}$.

Let u_F be the distribution on $X_F \times \mathcal{O}_F$ represented by the smooth density

$$h_F(a) |da| |dt|.$$

Its B-function is of $\mathcal{C}^{\text{exp}}$ -class by closure under parameterized integration, so $(u_F)_F$ is a smooth $\mathcal{C}^{\text{exp}}$ -class family in the sense of [1]. Consider the restriction of the polynomial mapping

$$\begin{aligned} f_F : X_F \times \mathcal{O}_F &\longrightarrow X_F \times F, \\ (a, t) &\longmapsto (a, t^2). \end{aligned}$$

Theorem 1. *For every F as above, f_F is proper on $\text{Supp}(u_F)$, and*

$$\text{WF}_{F^\times}(f_{F,*}u_F) = \{((a, 0), (0, 0, \eta)) : a \in X_F, h_F(a) \neq 0, \eta \in F^\times\}. \quad (1)$$

In particular, this wave-front set is not a definable subset of the cotangent bundle. Thus polynomial pushforward of a smooth $\mathcal{C}^{\text{exp}}$ -class distribution need not have definable wave-front set.

3 Proof intuition

The fold $t \mapsto t^2$ has one critical value, zero. Its pushforward of Haar measure is smooth away from zero but has ball masses $q_F^{-\lceil r/2 \rceil}$ at zero, which is incompatible with any locally constant density there. Multiplying the input by a locally constant coefficient $h_F(a)$ switches that fold singularity on exactly where $h_F(a) \neq 0$. The source paper's pushforward theorem forces every wave-front covector into the vertical conormal line of the fold, and its singular-support theorem forces this line to occur wherever the coefficient is nonzero. Finally, the switching locus is nondefinable because its complement encodes the exponential graph $n = q_F^m$ in the value group, where only Presburger sets are definable.

4 Proof

Lemma 1. *Let*

$$\nu_F = (t \mapsto t^2)_*(1_{\mathcal{O}_F}|dt|)$$

be the pushforward distribution on F . Then $\text{Sing}(\nu_F) = \{0\}$.

Proof. For $r \geq 0$, write $B_r(0) = \{s \in F : \text{ord } s \geq r\}$. Directly from the definition of pushforward,

$$\nu_F(B_r(0)) = \text{vol}\{t \in \mathcal{O}_F : \text{ord}(t^2) \geq r\} = q_F^{-\lceil r/2 \rceil}.$$

If ν_F were smooth near zero, it would be represented there by a locally constant function g . For some R , the function g would be constant, say equal to c , on $B_R(0)$. Consequently $\nu_F(B_r(0)) = cq_F^{-r}$ for every $r \geq R$, contradicting the displayed formula.

For $s_0 \neq 0$, the square map has no critical preimage: at any preimage t_0 , its derivative is $2t_0 \neq 0$. The non-archimedean inverse function theorem gives finitely many local inverse branches, and change of variables represents ν_F by a locally constant density near s_0 . If s_0 has no preimage in $\mathcal{O}_F$, the distribution vanishes on a sufficiently small neighborhood of s_0 , since the nonzero squares form a clopen subset of $F^\times$. Hence ν_F is smooth away from zero. $\square$

Lemma 2. *For $v_F = f_{F,*}u_F$,*

$$\text{Sing}(v_F) = \{(a, 0) : a \in X_F, h_F(a) \neq 0\}.$$

Proof. For a test function φ on $X_F \times F$, Fubini's theorem gives

$$v_F(\varphi) = \int_{X_F} h_F(a) \int_{\mathcal{O}_F} \varphi(a, t^2) |dt| |da|.$$

Thus $v_F = (h_F|da|) \otimes \nu_F$. Both h_F and ν_F are smooth away from the claimed set, by local constancy of h_F and Lemma 1.

Fix a_0 with $h_F(a_0) \neq 0$. On a compact open neighborhood U of a_0 , the function h_F is a nonzero constant. If v_F were smooth near $(a_0, 0)$, test its locally constant representing density against a Schwartz–Bruhat function in the a -variable supported in U and having nonzero integral. The resulting distribution in the s -variable would be a nonzero scalar multiple of ν_F and would be smooth near zero, contrary to Lemma 1. Conversely, if $h_F(a_0) = 0$, local constancy gives a neighborhood U on which $h_F = 0$, and v_F vanishes on $U \times F$. This proves the equality. $\square$

Proof of Theorem 1. First, the support of u_F is

$$\{a \in X_F : h_F(a) \neq 0\} \times \mathcal{O}_F,$$

a closed subset of $X_F \times \mathcal{O}_F$. If $K \subset X_F \times F$ is compact, then $f_F^{-1}(K) \cap \text{Supp}(u_F)$ is a closed subset of the compact set $\text{pr}_{X_F}(K) \times \mathcal{O}_F$. Hence f_F is proper on the support, as required for the pushforward.

The input u_F is smooth, so its wave-front set is empty. The pushforward theorem [1, Theorem 2.9.6] therefore yields

$$\text{WF}_{F^\times}(v_F) \subset N_{f_F}.$$

The differential of f_F is

$$Df_F(a, t) = \text{diag}(1, 1, 2t).$$

It follows directly from the definition of N_{f_F} that

$$N_{f_F} = \{((a, 0), (0, 0, \eta)) : a \in X_F, \eta \in F^\times\}.$$

Indeed, a nonzero target covector is killed by the transpose differential only at $t = 0$, and then only its last component can be nonzero.

The projection of wave-front set to the base equals singular support by [1, Proposition 2.8.3]. Lemma 2 therefore implies that, for each a with $h_F(a) \neq 0$, at least one point of $\text{WF}_{F^\times}(v_F)$ lies over $(a, 0)$. The preceding upper bound makes that covector $(0, 0, \eta_0)$ for some $\eta_0 \neq 0$. Since this is the $F^\times$ -wave-front set, conicity under multiplication by $F^\times$ then supplies every $(0, 0, \eta)$, $\eta \neq 0$. There is no wave front over $h_F(a) = 0$, again by Lemma 2. This proves (1).

It remains to prove nondefinability. Let

$$Z_F = \{(x, y) \in X_F : h_F(x, y) = 0\}.$$

Its image under the definable valuation map is

$$\text{ord}(Z_F) = \{(m, n) \in \mathbb{N}^2 : n = q_F^m\}.$$

If Z_F were definable, this image would be definable in the induced value group structure. By valued-field quantifier elimination in the generalized Denef–Pas setting, that induced structure is Presburger. But a Presburger-definable function on $\mathbb{N}$ is eventually affine on each class modulo some fixed integer (a standard consequence of Presburger cell decomposition; see [2]), whereas $m \mapsto q_F^m$ is exponential. Thus Z_F , and hence its complement $\{h_F \neq 0\}$, is not definable.

Finally, if the wave-front set in (1) were definable, then so would be its definable slice

$$\{a \in X_F : ((a, 0), (0, 0, 1)) \in \text{WF}_{F^\times}(v_F)\} = \{a \in X_F : h_F(a) \neq 0\},$$

a contradiction. The theorem follows. $\square$

5 Verification notes

The proof uses no numerical or symbolic computation. The following checks are independent and exact.

- The domain and target are definable open submanifolds, and f_F is the restriction of a polynomial map.
- The density is locally constant and constructible, so the input is both smooth and of $\mathcal{C}^{\text{exp}}$ -class.
- Properness is checked on the actual support, not assumed globally.
- The singularity of the fold measure is witnessed by its exact ball masses; no asymptotic estimate is used.
- The exact wave-front equality combines two source results: the pushforward upper bound and equality of projected wave front with singular support.
- Nondefinability is reduced to the standard one-variable structure of Presburger-definable functions.

6 Scope and interpretation

The density h_F is a legitimate smooth constructible/ $\mathcal{C}^{\text{exp}}$ -class density, but its zero locus is not definable. The construction therefore disproves a blanket claim that polynomial pushforwards of smooth $\mathcal{C}^{\text{exp}}$ -class distributions always have definable wave-front sets. It does not decide whether a narrower positive theorem holds when the input density has definable zero and nonzero loci, or when it is a fixed algebraic measure. Nor does it characterize all wave-front sets arising from polynomial pushforwards.

7 Novelty check

On July 22, 2026, a bounded search covered the run’s registry and lightweight indexes, the local arXiv source corpus, the follow-up arXiv:1909.09048, and arXiv web searches combining the source id and exact wording with “definable”, “polynomial push-forward”, “smooth distribution”, and “non-archimedean wave front set”. It located [1], Raibaut’s work on motivic wave-front sets, and the later WF-holonomicity result, but no matching counterexample or theorem. This establishes only bounded-search novelty confidence; it is not an exhaustive priority claim.

8 Human-review recommendation

The packet is recommended for expert review as a likely valid counterexample. The principal interpretive question is whether the authors intended an unstated narrower class of smooth densities with definable zero locus. Under the broad $\mathcal{C}^{\text{exp}}$ -class reading provided by the paper’s context, every hypothesis is explicit and the conclusion fails exactly as in (1).

References

- [1] R. Cluckers, I. Halupczok, F. Loeser, and M. Raibaut, *Distributions and wave front sets in the uniform non-archimedean setting*, arXiv:1706.03003, 2017 (version used: 2021).
- [2] R. Cluckers, *Presburger sets and p -minimal fields*, Journal of Symbolic Logic 68 (2003), 153–162; arXiv:math/0206197.